



Swiggy Sales & Customer Insights Analysis (January–August 2025)

Identifying Growth Opportunities Through Data Analytics

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Tools & Technologies: MySQL, PowerBI, Microsoft PowerPoint, Excel

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Table Of Contents

1. Executive Summary	2
2. Business Problem & Objectives	4
3. Dataset Overview	6
4. Data Preparation & Data Model	8
5. KPI Overview	11
6. Business Analysis & Key Findings	12
6.1 Demand Trends	12
6.2 City Performance	15
6.3 Restaurant & Catgory Performance	17
6.4 Special Case Observations and their findings.....	19
6.4 Customer Experience	22
6.5 Why Does Gujarat Receive More 5-Star Ratings?.....	23
7. Business Recommendations	24
8. Limitations	26
9. Conclusion	28
10. Appendix	29

Executive Summary

Objective

The objective of this project was to analyze Swiggy's order data from January to August 2025 to evaluate overall business performance, identify key demand patterns, understand city-and restaurant-level performance, and generate actionable recommendations to support business growth. The analysis focused on answering business questions related to order trends, revenue generation, customer satisfaction, restaurant performance, and market expansion opportunities.

Dataset Overview

The analysis was performed on a dataset containing 197,403 food orders across multiple Indian cities over an eight-month period. The dataset included information such as order date, restaurant, location, food category, dish name, price, customer rating, and rating count. Data preparation involved handling missing values, removing duplicate records, standardizing inconsistent category names, and designing a star schema consisting of one fact table and multiple dimension tables to support efficient analysis.

Key Findings

- Swiggy processed approximately 197K orders and generated nearly ₹53 million in revenue during the analysis period while maintaining a strong average customer rating of 4.34.
- Order demand was strongest from Thursday through Sunday, highlighting clear opportunities for targeted weekend marketing campaigns.
- Bengaluru recorded the highest order volume, while Lucknow emerged as the second-highest revenue-generating city despite having comparatively fewer orders, primarily due to stronger contribution from premium quick-service restaurant chains.
- Jaipur demonstrated high order volume but comparatively lower revenue because of its lower average revenue per order.
- Gangtok, Kohima, and Aizawl showed limited restaurant availability and delivery coverage, indicating potential opportunities for strategic market expansion.
- Categories such as Recommended, Main Course, Burgers, Desserts, and Beverages consistently contributed the highest order volume and revenue.
- Customer satisfaction remained consistently strong, with only a very small proportion of orders receiving ratings below 3, indicating an overall positive customer experience.

Key Recommendations

- Launch targeted marketing campaigns between Thursday and Sunday to maximize demand during peak ordering periods.
- Expand restaurant partnerships and delivery coverage in underserved cities such as Gangtok, Kohima, and Aizawl.
- Analyze and replicate the operational and promotional strategies contributing to Lucknow's strong revenue performance in other major cities where applicable.
- Increase average order value in Jaipur by introducing premium meal bundles, combo offers, and higher-value menu options at high-performing local restaurants.
- Encourage cross-selling by recommending complementary categories such as Desserts and Beverages during checkout to improve average order value.
- Increase the visibility of highly rated products and restaurants to strengthen customer engagement and support sustainable revenue growth

Business Problem & Objectives

Business Problem

- Swiggy operates across numerous cities, restaurant partners, and food categories, generating a large volume of transactional data every day. While overall business performance can be measured through key metrics such as orders and revenue, identifying the underlying factors driving growth or limiting performance requires a structured analytical approach.
- This project was undertaken to transform raw order-level data into actionable business insights by identifying demand patterns, evaluating city and restaurant performance, understanding customer purchasing behaviour, and recommending opportunities to improve revenue.

Business Objectives

The analysis was designed to answer the following business questions:

- How did Swiggy perform overall in terms of orders, revenue, average dish price, and customer ratings?
- How did order volume and revenue change over time?
- Which weekdays generated the highest customer demand?
- Which cities contributed the highest and lowest order volume and revenue, are there any anomalies?
- Which restaurant chains and food categories contributed most to business performance?
- What insights can be derived from customer ratings and regional performance?
- What business actions can be recommended to improve growth, revenue, and customer satisfaction?

Scope of Analysis

The analysis covers food order transactions from January to August 2025, comprising 197,403 orders across multiple Indian cities.

The project focuses on:

- Sales performance analysis
- Revenue analysis
- Time-based demand analysis
- Geographic performance analysis
- Restaurant performance analysis
- Category performance analysis
- Customer rating analysis
- Business recommendation generation

The analysis is based solely on the available transactional dataset. Customer demographics, marketing campaign data, delivery partner performance, operational costs, and profitability metrics were not available and are therefore outside the scope of this study.

Dataset Overview

Dataset Source

The dataset used in this project is a simulated Swiggy food delivery dataset containing transactional order information collected across multiple Indian cities. The dataset represents customer food orders placed between January 2025 and August 2025 and is intended for analytical and learning purposes.

Dataset Time Period

- Start Date: January 2025
- End Date: August 2025
- Duration: 8 Months

Dataset Size

Attribute	Value
Total Records	197,403 Orders
Time Period	January–August 2025
Number of States	28
Number of Cities	28
Number of Restaurants	984
Number of Food Categories	4772
Number of Dishes	56,582

Data Granularity (Grain)

Each record in the dataset represents one food order placed by a customer for a specific dish from a particular restaurant at a given location and date.

This transaction-level granularity enables analysis across multiple business dimensions such as time, geography, restaurants, food categories, customer ratings, and revenue.

Key Attributes

The dataset contains the following business attributes:

Column	Description
Order Date	Date on which the order was placed
State	State where the order was placed
City	City where the Order was placed
Location	Delivery locality within the city
Restaurant Name	Restaurant fulfilling the order
Category	Food Category of the Ordered Dish
Dish Name	Name of the ordered dish
Price	Price of the Ordered Dish (INR)
Rating	Customer Rating for the dish / order
Rating Count	Number of Customer ratings received

id	state	city	order_date	restaurant_name	location	category	dish_name	price	rating	rating_count
1	Karnataka	Bengaluru	2025-06-29	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butter Murukku-200gm	133.90	4.0	0
2	Karnataka	Bengaluru	2025-04-03	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badam Milk	52.00	4.5	25
3	Karnataka	Bengaluru	2025-01-15	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chow Chow Bath	117.00	4.7	48
4	Karnataka	Bengaluru	2025-04-17	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesari Bath	65.00	4.6	65

Fig: Sample of the Swiggy transaction dataset

The dataset captures multiple business dimensions, including geography, restaurants, products, pricing, and customer feedback. This makes it suitable for analyzing operational performance, customer demand patterns, revenue generation, and identifying opportunities for business growth.

Data Cleaning & Preparation

Before performing any business analysis, the dataset was cleaned and transformed to improve data quality, ensure consistency, and support efficient analytical reporting. The preparation process focused on eliminating data quality issues and designing a structured data model suitable for analysis.

Data Cleaning Activities

1. Missing Value Assessment

The dataset was examined for missing values across all business attributes to ensure that key analytical fields such as order date, restaurant, category, price, and ratings contained valid information before analysis.

2. Duplicate Record Removal

Duplicate transactions were identified using SQL window functions and removed to avoid inflated order counts and revenue calculations. This step ensured that every transaction represented a unique customer order.

```
-- Finding the duplicate rows
with duplicates_rn as (
  select *, row_number() over (partition by state,
    city,
    order_date,
    restaurant_name,
    location,
    category,
    dish_name,
    price,
    rating,
    rating_count
  ) as rn
  from swiggy_data )
select * from duplicates_rn where rn>1;
-- We got 27 rows which were duplicate
```

Fig: Example SQL used during duplicate detection

3. Data Standardization

Several fields contained inconsistent naming conventions and formatting. Data was standardized by:

- Correcting inconsistent category names
- Ensuring consistent date formats
- Validating numerical fields such as price and ratings

These transformations improved data consistency and reduced analytical errors

```
-- Correcting inconsistent category  
|  
select count(*) from swiggy_data where category like 'Acc%' and category like '%ts'; -- 517  
select * from swiggy_data where category = "Accompaninents";  
select count(*) from swiggy_data where category = "Accompaniments"; -- 514 , means those are main three needs to be updated  
update swiggy_data  
set category = "Accompaniments"  
where category = "Accompaninents";  
  
select count(*) from swiggy_data where category = "Accompaniments"; -- 517 , now they are updated
```

Fig: Example SQL used during correcting category inconsistency

4. Data Validation

Following the cleaning process, the dataset was validated by verifying record counts, checking for duplicate entries, and ensuring that there is no data loss.

Data Modeling

To improve query performance and simplify analytical reporting, the cleaned transactional dataset was transformed into a Star Schema.

The model consists of:

Fact Table

- **fact_swiggy_orders**
 - Stores transactional measures such as price, rating, rating count, and foreign keys linking to each business dimension.

Dimension Tables

- dim_date
- dim_location
- dim_restaurant
- dim_category
- dim_dish

This dimensional model separates descriptive attributes from transactional data, enabling efficient aggregation, simplified SQL queries, and optimized reporting in Power BI. The resulting star schema served as the foundation for all SQL analysis, business insights, and dashboard development presented in this report.

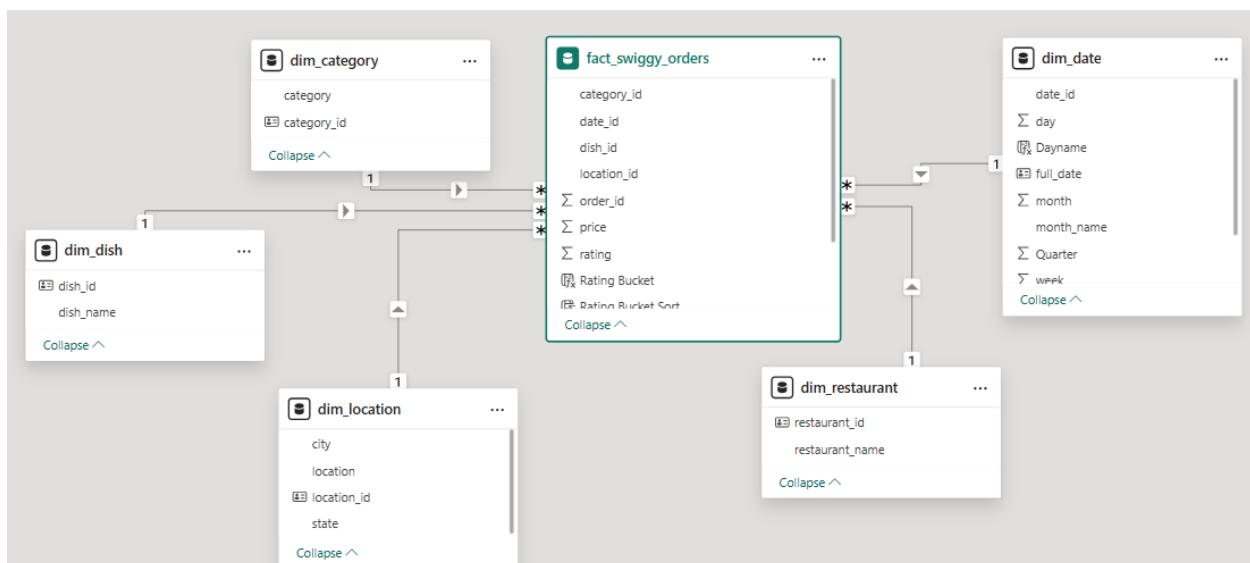


Fig: Star schema implemented for analytical reporting

KPI Overview

The dashboard begins with four key performance indicators (KPIs) that provide an overall view of Swiggy's business performance during the analysis period (January–August 2025). These metrics serve as the foundation for the detailed analyses presented in the following sections.



Fig: Overall business KPIs.

Business Analysis & Findings

Demand Trends

Monthly Trends

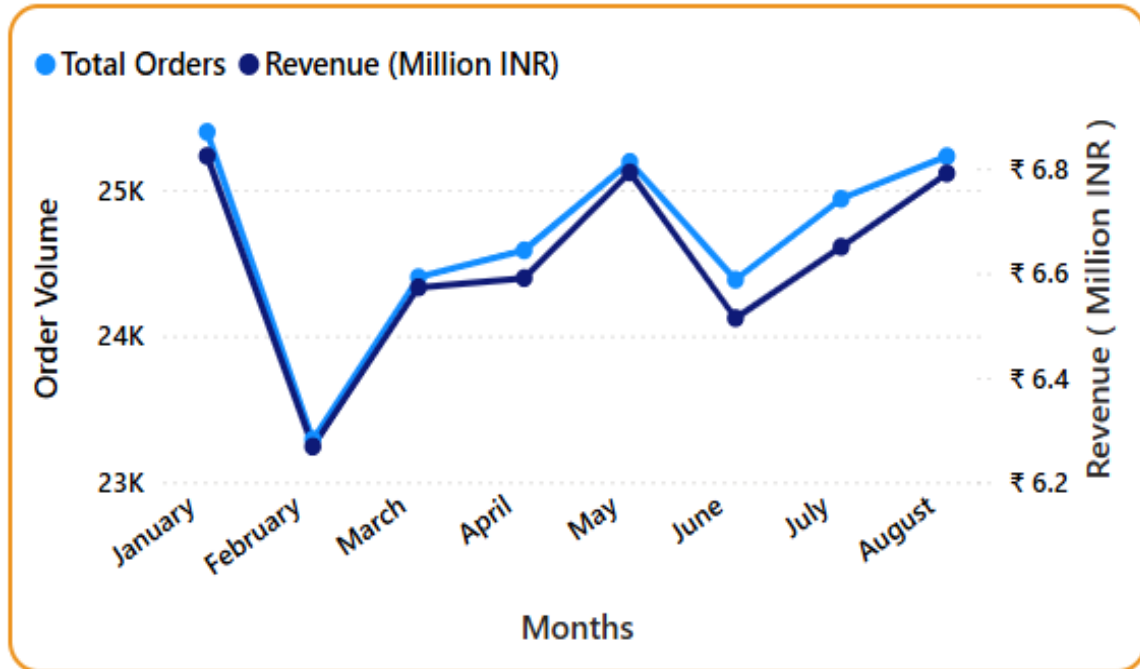


Fig: Monthly Trend of Order and Monthly Revenue generated

Observations

- Orders peaked in January, May, and August.
- Revenue closely followed order trends.
- February, March, and June recorded relatively lower activity and were investigated further.

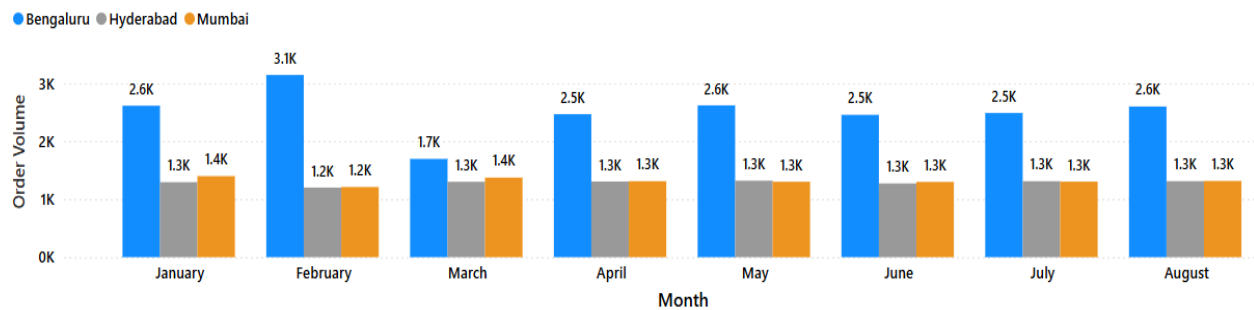


Fig: Comparing Monthly Order volume of Bangalore, Mumbai, Hyderabad

Interpretations

- The decline in order volume was relatively small rather than a significant business downturn.
- February's lower orders are largely explained by having fewer calendar days.
- March's decline was primarily associated with reduced order volume from Bengaluru.
- June experienced weaker weekend demand during Weeks 25–26.

Recommendations

- Investigate the factors contributing to the lower order volume in Bengaluru during March, including restaurant participation, promotional campaigns, and operational performance, to determine whether targeted interventions are required.
- Review operational and marketing activities during Weeks 25–26 in June to understand the decline in weekend orders and identify opportunities to restore demand during peak periods.

Weekly Trends

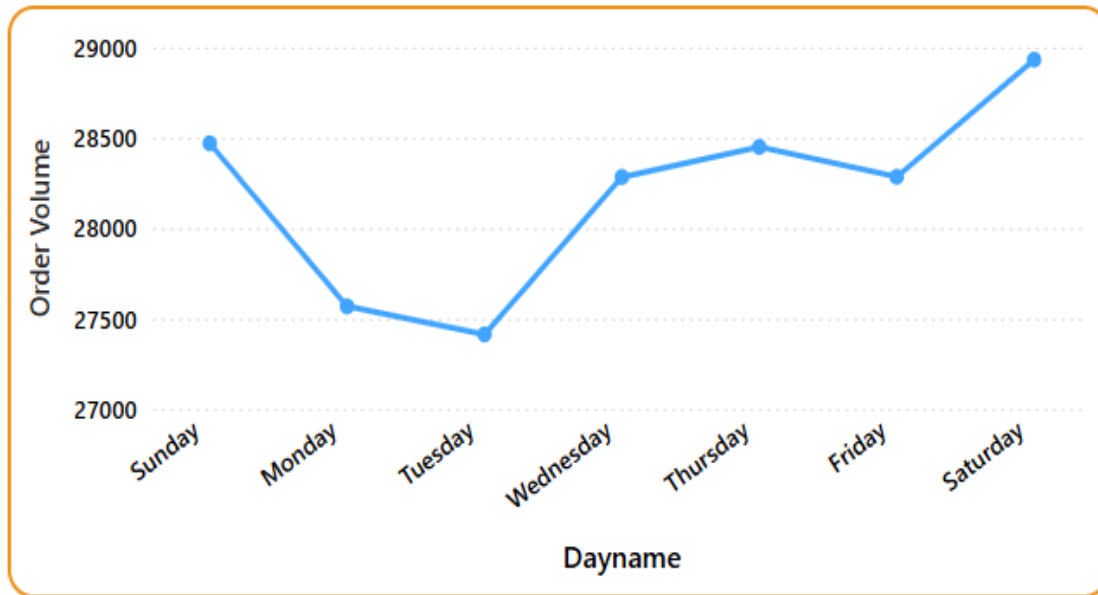


Fig: Weekly Order Trend

Observations

- Customer demand increases steadily from Thursday and reaches its highest levels on Saturday and Sunday, indicating a clear weekend ordering pattern.

Recommendations

- This creates opportunities for targeted weekend marketing and promotional campaigns.

City Performance

Top Performing Cities

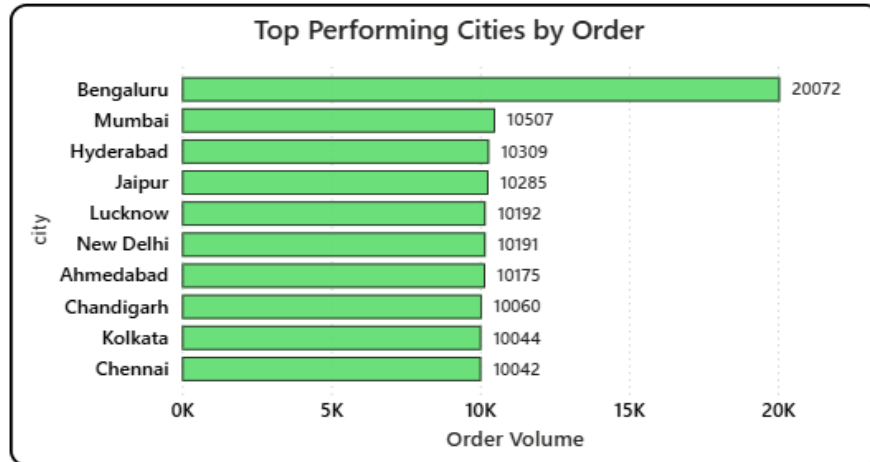


Fig: Top Performing Cities by Order

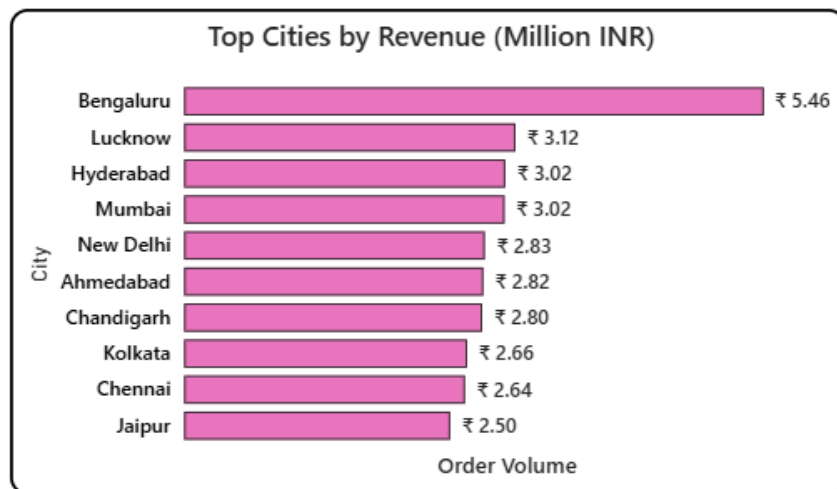


Fig: Top Performing Cities by Revenue

Observations

- Bengaluru, Mumbai, and Hyderabad generate the highest order volumes.
- While Bengaluru also leads in revenue, Lucknow ranks 2nd in revenue despite having only the 5th highest order volume.
- Jaipur ranks 4th in order volume but falls to 10th in revenue, indicating lower revenue generated per order. We have investigated these cases further.

Expansion Opportunities

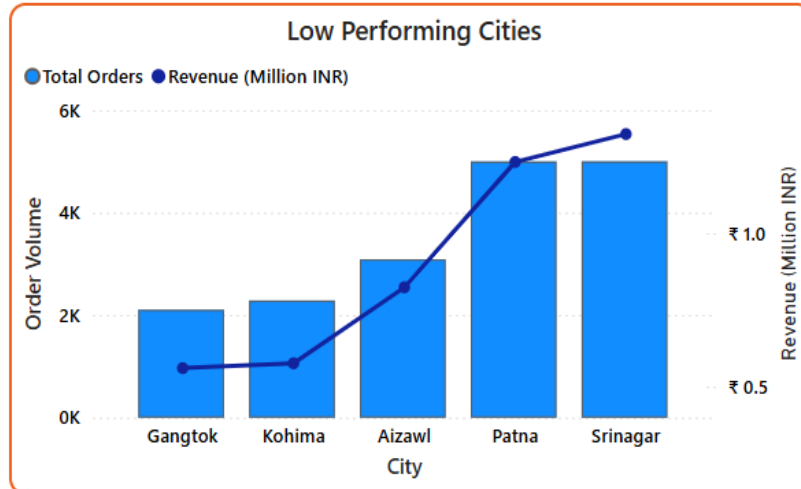


Fig: Low Performing Cities by Order volume and Revenue Generation

Observations

- Cities with low order volume and low revenue - Gangtok, Kohima, Aizawl.

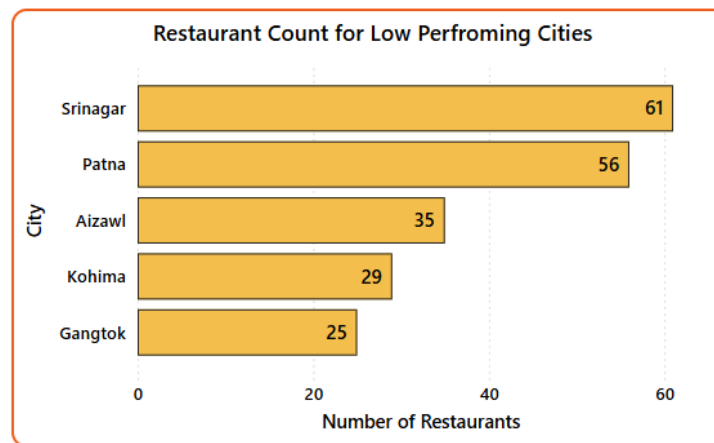


Fig: Restaurant Count for Low Performing cities

Interpretations

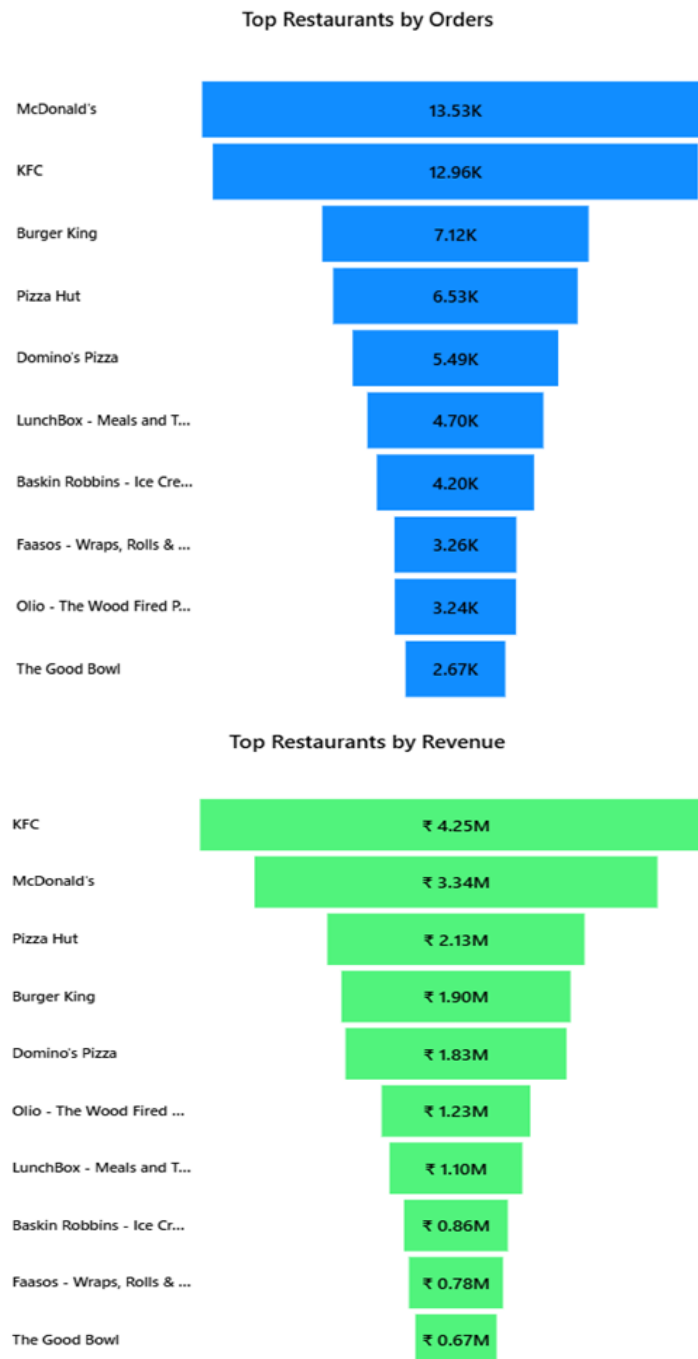
- All three cities have a relatively small restaurant network (25–35 restaurants) and limited delivery coverage (7–11 locations), which is likely associated with lower order volumes and revenue.

Recommendation

- Expanding restaurant partnerships and delivery coverage could improve market penetration in these cities.

Restaurant & Category Performance

Top Restaurant Chains



Top-performing restaurant chains appear to benefit from strong demand across multiple high-performing categories such as Burgers, Desserts, and Beverages, Pizza which is likely associated with their higher overall order volume and revenue.

High Performing Categories

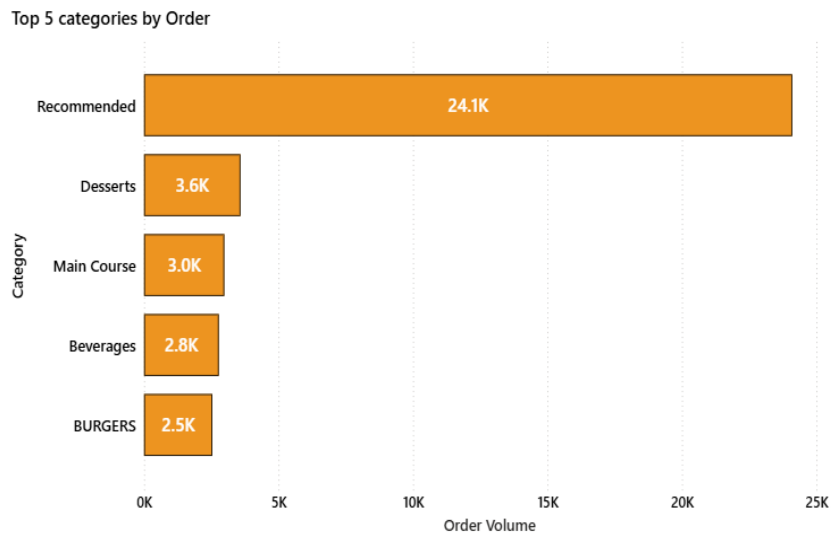


Fig: Top 5 categories by Order

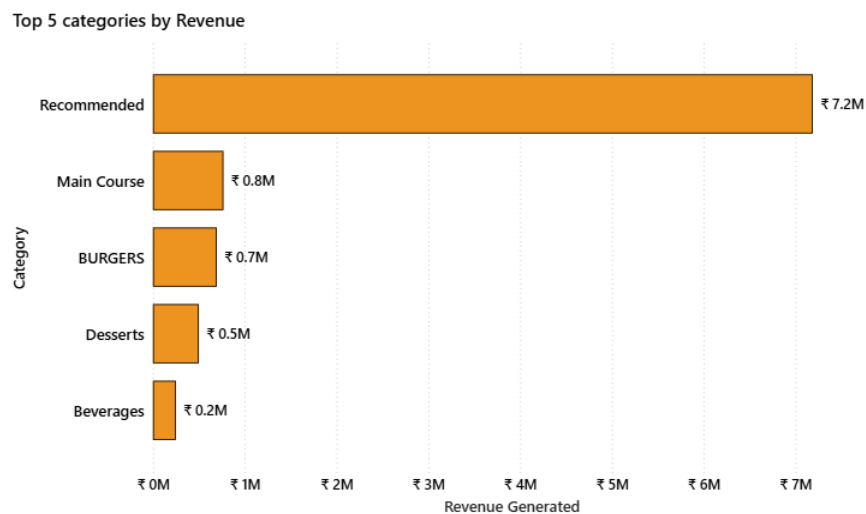


Fig: Top 5 categories by Revenue

Observations

- Recommended, Main course, Burgers, Desserts, Beverages are giving high total revenue.
- Recommended, Desserts, Main Course, Beverages, Burgers are having top order count

Special Case Observations and their findings

Why Does Lucknow Generate Higher Revenue Than Mumbai and Hyderabad?

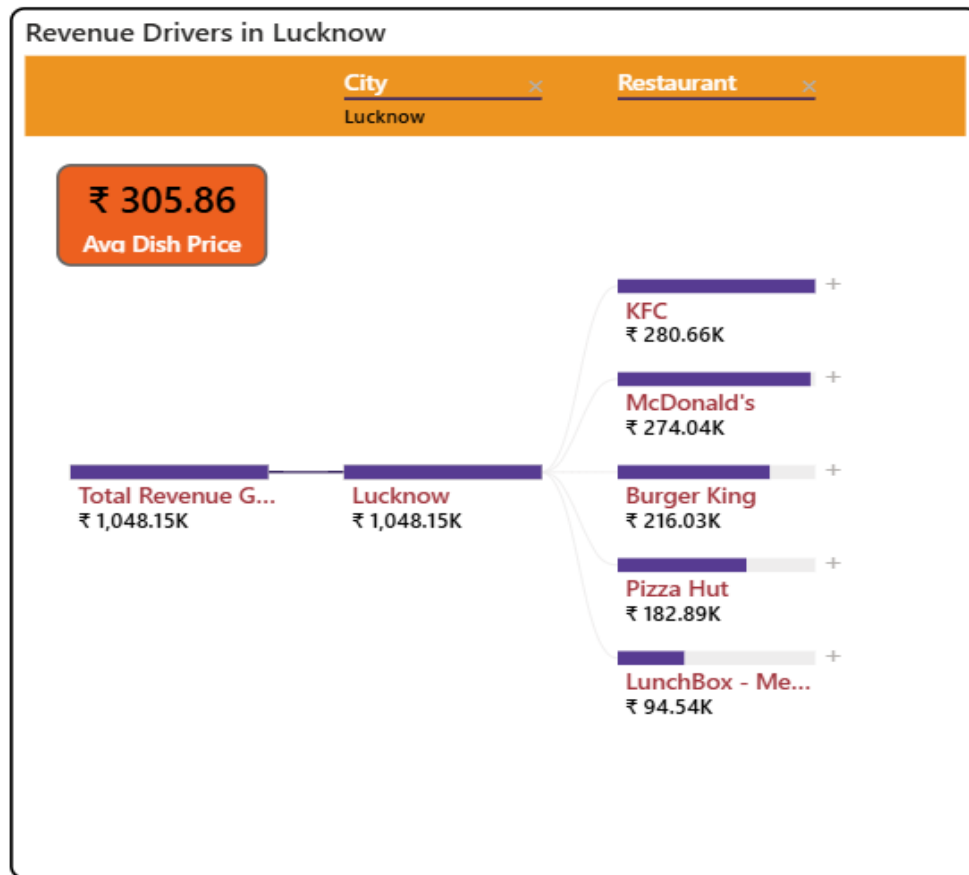


Fig: Decomposition Tree of Revenue generation in Lucknow by top 5 restaurants

Observations

- Lucknow has one of the highest average dish prices among the top cities.
- A significant share of orders comes from premium quick-service chains such as KFC, McDonald's, Burger King and Pizza Hut.

Interpretations

- This combination of High Average Order value and QSR chains is likely associated with higher revenue generation despite lower order volume.

Why Jaipur generates low revenue despite of high order count?

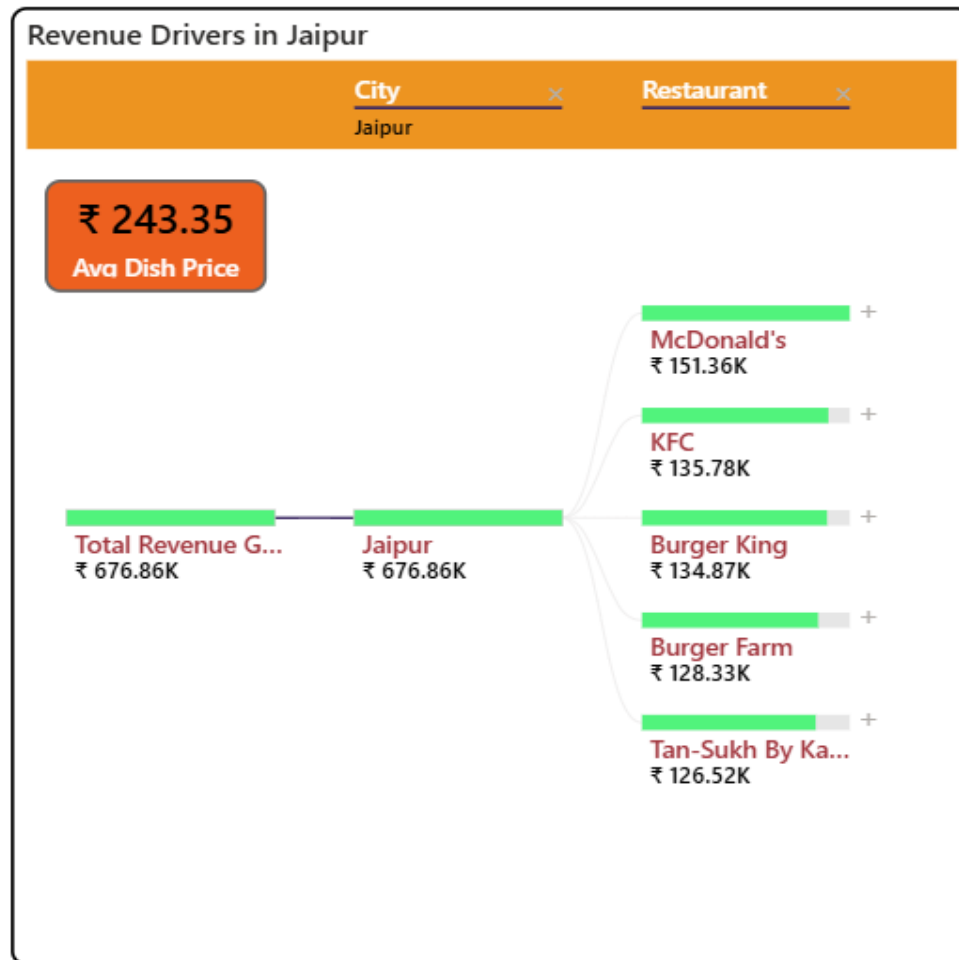


Fig: Decomposition Tree of Revenue generation in Jaipur by top 5 restaurants

Observations

- Jaipur's average dish price (₹243) is lower than other leading cities, reducing revenue generated per order.
- A large share of orders comes from local restaurants such as Burger Farm and Tan-Sukh By Kanha, which have lower average menu prices than premium chains.

Interpretations

- Introducing premium meal bundles or higher-value menu options at high-volume restaurants could increase average order value while maintaining demand.

Main Course Demand Is Distributed Across Many Restaurants

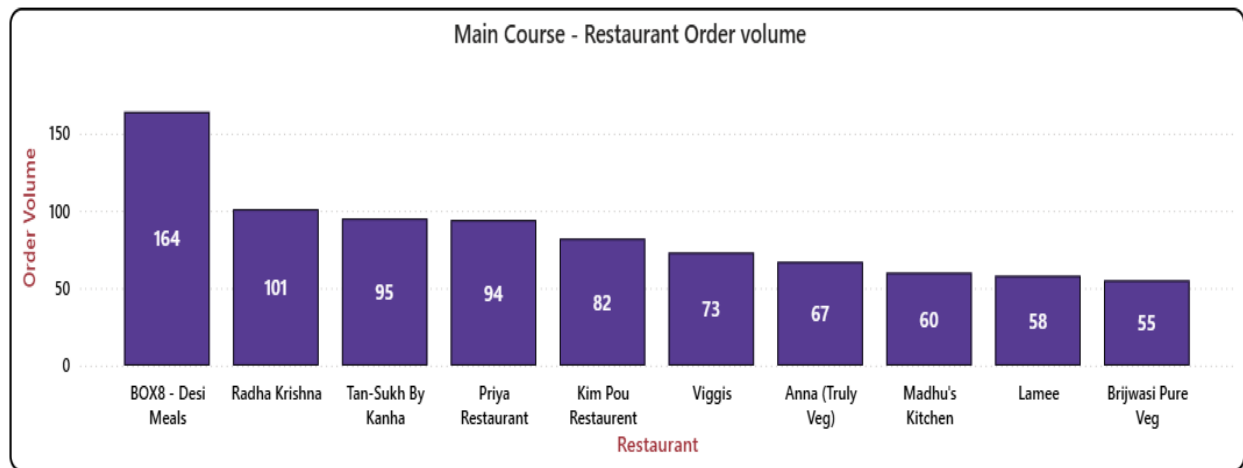


Fig: Restaurant Wise Distribution of Order Volume for Main Course Category

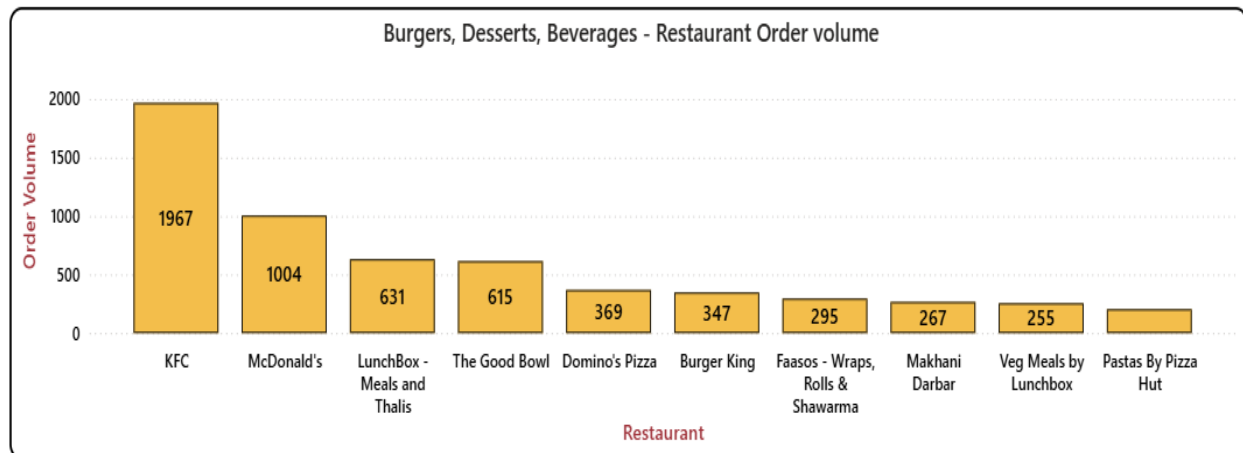


Fig: Restaurant Wise Distribution of Order Volume for Burgers, Desserts, Beverages Categories

Observations

- Unlike categories such as Burgers or Desserts, where demand is concentrated among a few major restaurant chains, Main Course orders are distributed across a wider range of restaurants.
- This broader distribution reduces the likelihood of any single restaurant dominating overall order volume or revenue for Main Course category.

Customer Experience

Rating Distribution

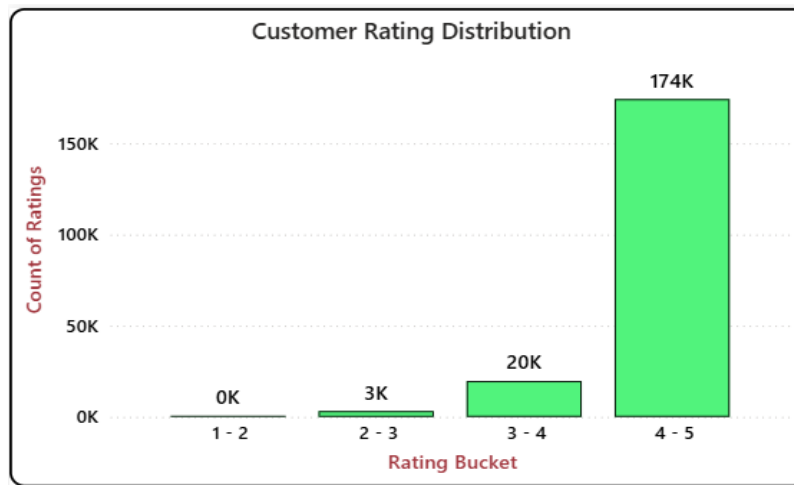


Fig: Customer Rating Distribution

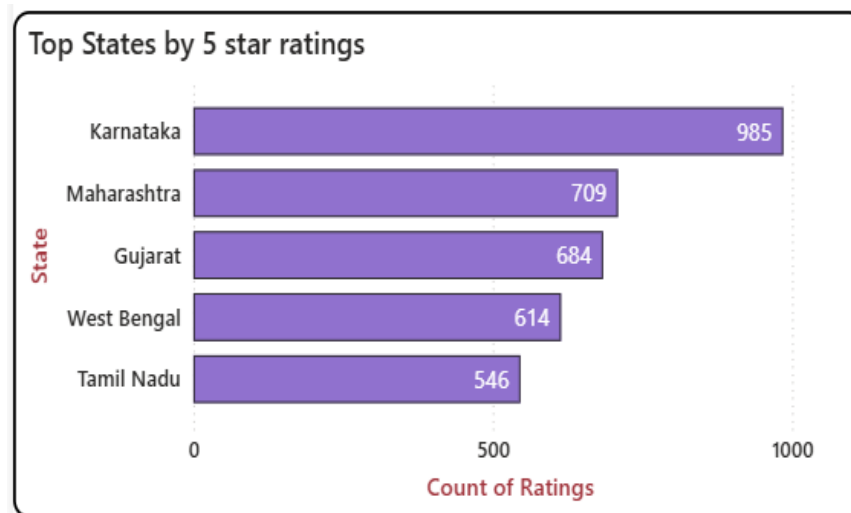


Fig: Top States by 5 star ratings

Observations

- 4.40 is the rating which has received the highest rating count , nearly 43.3 % of total orders, indicating a consistently positive customer experience
- Only 1.7% of orders received ratings below 3, suggesting relatively few poor customer experiences.
- One interesting exception was Gujarat, which recorded unusually high 5-star ratings despite not being among the top-order states. This was investigated further.

Why Does Gujarat Receive More 5-Star Ratings?

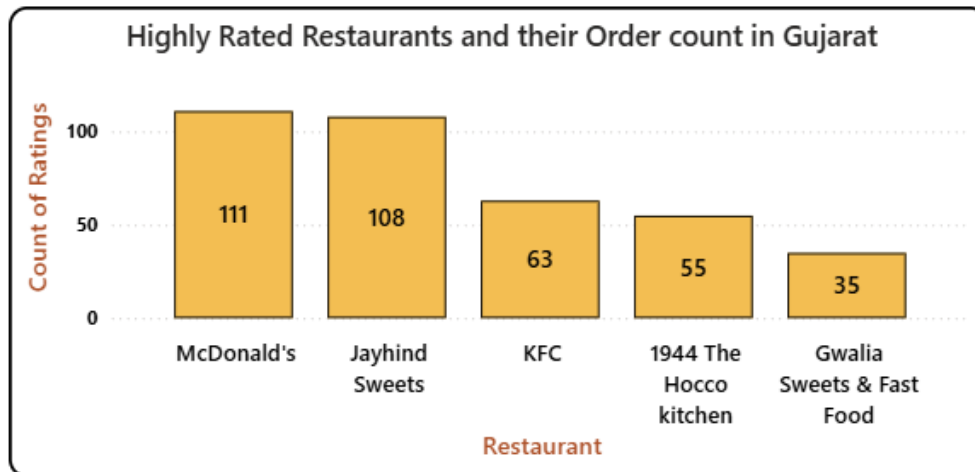


Fig: Highly Rated Restaurants and their Order Count in Gujarat

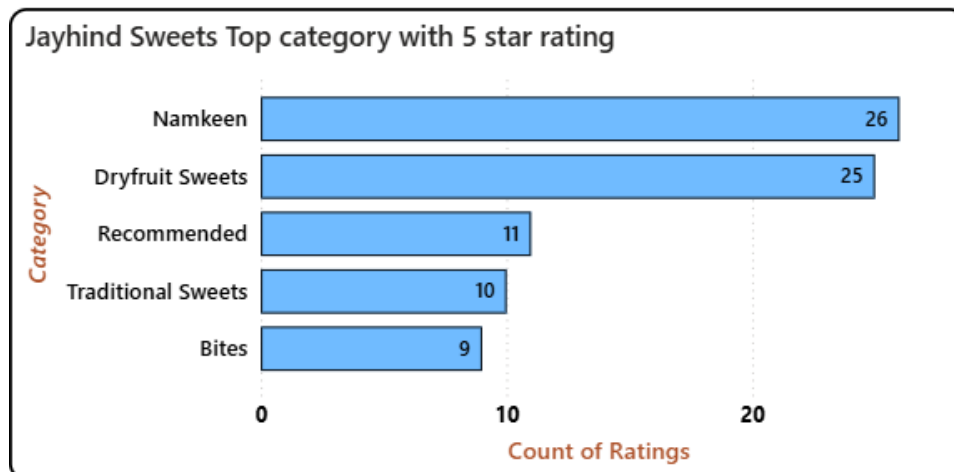


Fig: Highly Rated Restaurants and their Order Count in Gujarat

Observations

- Gujarat records a disproportionately high number of 5-star ratings despite relatively lower order volume.
- Within Gujarat, Jayhind Sweets stands out with approximately 19.5% of its orders receiving a 5-star rating.

Interpretations

- Namkeen and Dry fruit Sweets combine high demand with consistently high customer ratings, making them strong candidates for promotional campaigns.
- Swiggy could test increased visibility for these products through featured placements or recommendations and monitor the impact on order volume.

Business Recommendations

Based on the analyses conducted throughout this project, the following strategic recommendations are proposed to support business growth, improve operational performance, and enhance customer experience.

Marketing & Expansion

1. Leverage Weekend Demand

Order volume consistently increases from Thursday through Sunday, making weekends the strongest demand period. Swiggy should prioritize targeted marketing campaigns, promotional offers, and restaurant participation during these peak days to maximize order volume.

2. Expand Restaurant Network in Low Performing Cities

Cities such as Gangtok, Kohima, and Aizawl have comparatively fewer restaurants and limited delivery coverage. This analysis suggests an opportunity to expand restaurant partnerships and increasing service availability in these markets which would improve both order volume and revenue.

3. Replicate High-Performing City Strategies

Lucknow demonstrated strong revenue performance despite comparatively fewer orders. Swiggy should investigate the operational, promotional, and restaurant partnership strategies contributing to this performance and evaluate whether similar approaches can be adopted in other major cities.

Revenue Growth

1. Increase Average Order Value in Jaipur

Although Jaipur generates high order volume, its average revenue per order is comparatively lower. Encouraging premium meal bundles, combo offers, and higher-value menu items at high-performing local restaurants may improve overall revenue while maintaining existing demand.

2. Encourage Cross-Selling

Restaurants specializing in Main Course items can expand into complementary high-demand categories such as Desserts and Beverages. Additionally, Swiggy can consider recommending these items during checkout to increase average order value.

3. Promote High-Value Restaurant Categories

Top-performing restaurant chains consistently generate revenue from multiple high-performing categories. Identifying and promoting similar category combinations across other restaurants may contribute to higher revenue generation.

Customer Experience

1. Promote Highly Rated Products

Products that consistently receive strong customer ratings—such as Jayhind Sweets' Namkeen and Dryfruit Sweets in Gujarat—could be featured more prominently within the platform to increase customer engagement and evaluate their impact on order volume and revenue.

2. Monitor Customer Satisfaction

The overall customer rating remains consistently high throughout the analysis period. Swiggy should continue monitoring customer feedback and investigate restaurants receiving lower ratings to maintain service quality and customer satisfaction.

Limitations

While the analysis provides valuable business insights, the findings should be interpreted in the context of the available data and the following limitations:

1. The dataset covers only eight months (January–August 2025) and may not capture full-year seasonal patterns or long-term business trends.
2. The analysis is based solely on transactional order data. Customer demographic information (such as age, gender, or customer segments) was not available and therefore could not be incorporated into the analysis.
3. Information related to marketing campaigns, promotional offers, discounts, and advertising activities was not available. Consequently, fluctuations in demand could not be directly linked to marketing initiatives.
4. The dataset does not include operational metrics such as delivery time, restaurant preparation time, cancellation rate, or delivery partner performance, limiting the ability to investigate operational causes behind demand variations.

5. The analysis focuses on revenue and does not include cost, profitability, commission, or margin data. As a result, recommendations are intended to improve revenue and customer engagement rather than optimize profitability.
6. The dataset represents customer orders only and does not include competitor activity, market conditions, festivals, weather, or other external factors that may influence customer demand.
7. The recommendations presented in this report are based on the available transactional data and should be validated using additional operational and business data before implementation.

The analysis identifies relationships and patterns within the data but does not establish direct cause-and-effect relationships. Additional operational, marketing, and customer-level data would be required to validate the underlying drivers of observed trends.

Conclusion

This project demonstrates how SQL and Power BI can be used to transform raw transactional data into actionable business insights. Through data cleaning, dimensional modeling, exploratory analysis, and dashboard development, several opportunities related to revenue growth, market expansion, customer behavior, and restaurant performance were identified. While the analysis is constrained by the available dataset, the findings provide a structured foundation for further investigation and data-driven decision-making.

Appendix

Quarterly Trends

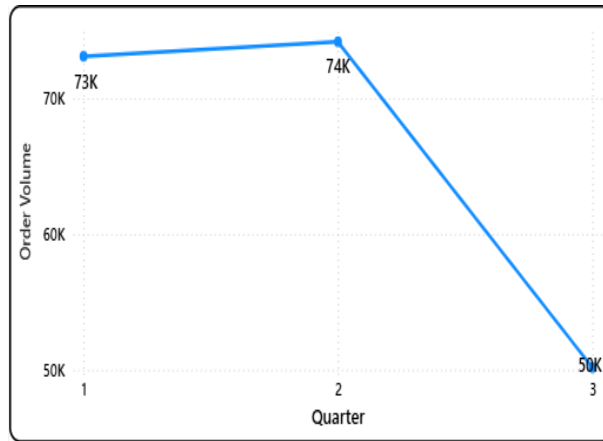


Fig: Quarterly Order Trend

- Quarterly trend for Order volume and revenue is same :
Q2 (April -May-June) is top performing,
Q1 (Jan-Feb-March) is second highest,
Q3 (July-Aug) is the 3rd Highest as it contained two months data only.

Most Ordered Dishes

Top 10 Dishes by Order		
Dish_name	Total Orders	Total Revenue Generated
Veg Fried Rice	321	₹ 57,765.70
Choco Lava Cake	303	₹ 32,942.35
Jeera Rice	265	₹ 44,505.33
Paneer Butter Masala	262	₹ 76,181.82
French Fries	248	₹ 35,218.02
Chicken Sausage	230	₹ 51,592.00
Chicken Fried Rice	228	₹ 48,594.82
Butter Naan	218	₹ 16,577.05
Margherita Pizza	203	₹ 37,220.00
Green Salad	197	₹ 22,355.32

Fig: Highly Ordered Dishes

Low Performing Restaurants

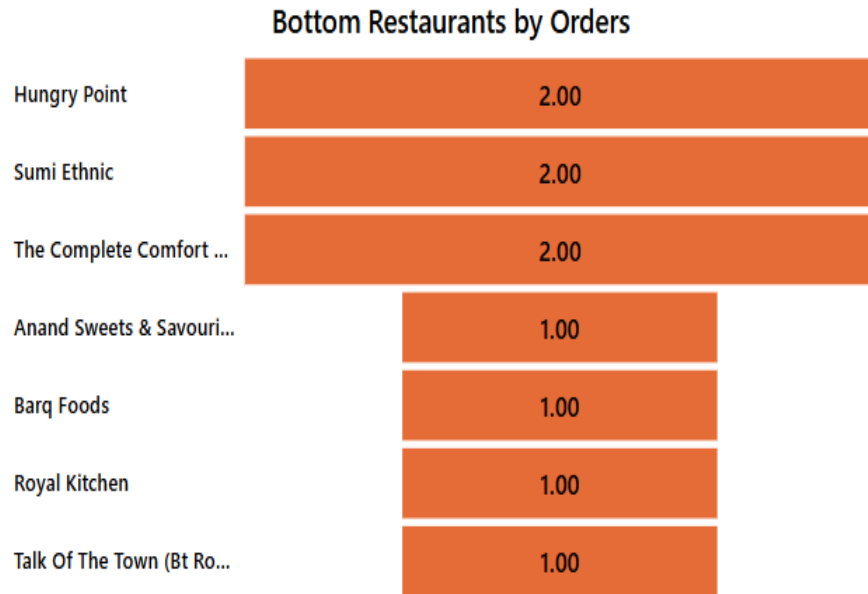


Fig: Low Order Volume Restaurant

Bottom Restaurants by Revenue

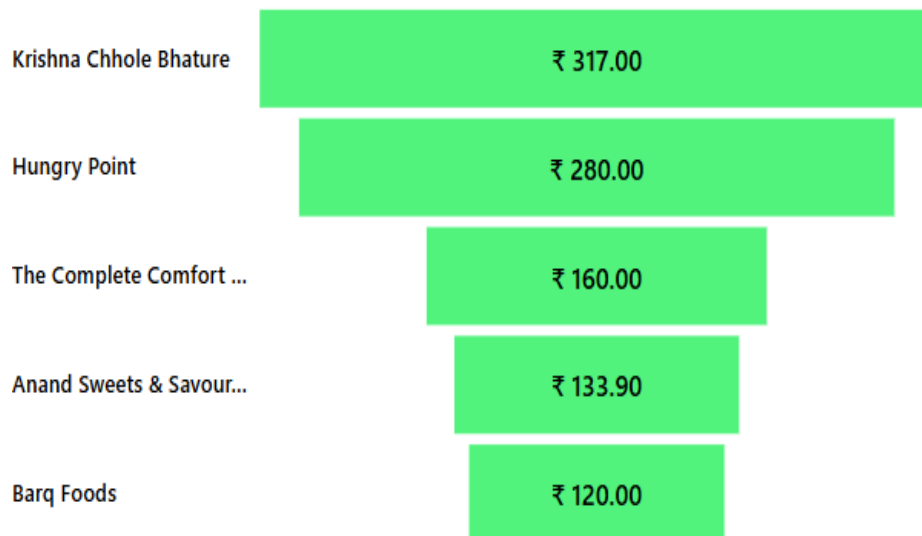


Fig: Low Revenue Generated Restaurants

Customer Spending Insights

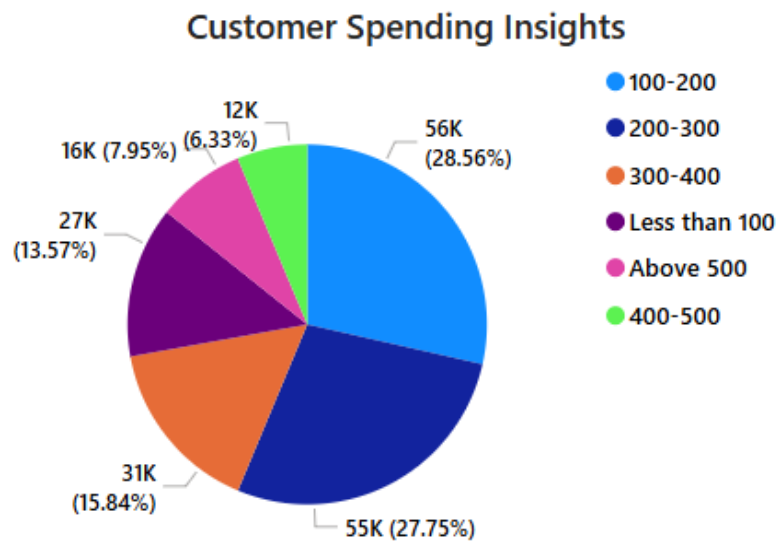


Fig: Customer Spending Insights

Observations

- We can see order are mostly coming from price bucket between 100-400 price bucket.
- Comparatively few orders are seen in 'under 100', '400-500' and '500 + category'

Additional SQL Queries for Reference

SQL for Dimensions and Fact tables

```
-- date table
use swiggy_database;
create table dim_date ( date_id int auto_increment primary key ,
    full_date date ,
    year int ,
    month int,
    month_name varchar(50) ,
    Quarter int,
    day int,
    week int);

select * from swiggy_data limit 5;

-- dim_location table
create table dim_location(location_id int auto_increment primary key , state varchar(100) , city varchar(100) , location varchar(250));

-- dim_restaurant table
create table dim_restaurant(restaurant_id int auto_increment primary key,restaurant_name varchar(300));

-- dim_category table
create table dim_category(category_id int auto_increment primary key, category varchar(300) );

-- dim_dish table
create table dim_dish( dish_id int auto_increment primary key , dish_name varchar(350));
```

Fig: SQL Query for dimension tables

```
-- let us now create fact table
use swiggy_database ;
select * from swiggy_data limit 5 ;
create table fact_swiggy_orders(
    order_id int auto_increment primary key,

    date_id int ,
    price numeric(10,2),
    rating numeric (4,2),
    rating_count int ,

    location_id int,
    restaurant_id int,
    category_id int,
    dish_id int,

    foreign key(date_id) references dim_date(date_id),
    foreign key(location_id) references dim_location(location_id),
    foreign key(restaurant_id) references dim_restaurant(restaurant_id),
    foreign key(category_id) references dim_category(category_id),
    foreign key(dish_id) references dim_dish(dish_id)
);
```

Fig: SQL Query for Fact table

SQL Analysis Work

Sample SQL queries of the performed analysis are attached below

```
-- Date Based Analysis
-- Monthly order trends and Monthly Revenue trend
select
d.year,
d.month,
d.month_name,
count(f.order_id) as Total_Orders_Monthly ,
concat(format((sum(f.price)/1000000),2),' Million INR') as Total_Revenue_Monthly
from fact_swiggy_orders f
join dim_date d
on f.date_id=d.date_id
group by d.year,
d.month,
d.month_name
order by Total_Orders_Monthly desc;

-- Jan , August , May had highest number of orders ,and same pattern is observed for total_revenue generated
-- Feb , March , June had lowest orders and same pattern is observed for total_revenue generated
```

Fig: SQL Query for Date Based Analysis

```
-- Top 10 cities by total_revenue

select
l.city,
concat(format(sum(f.price)/1000000,2),' Million INR') as total_revenue_city
from fact_swiggy_orders f
join dim_location l
on f.location_id = l.location_id
group by l.city
order by total_revenue_city desc
limit 10;
```

Fig: SQL Query for City Performance Analysis

```
-- Why Jaipur beign 4th highest in orders is giving revenue 10 th highest ?
with join_cte1 as
(select price, rating, rating_count,full_date,year,month,month_name,quarter, day,week,state,city, location, restaurant_name,
category,dish_name
from fact_swiggy_orders f
join dim_date d on f.date_id=d.date_id
join dim_location l on f.location_id=l.location_id
join dim_restaurant r on f.restaurant_id=r.restaurant_id
join dim_category c on f.category_id =c.category_id
join dim_dish dd on f.dish_id=dd.dish_id
order by order_id )
select restaurant_name,count(*) from join_cte1 where city ='Jaipur'
group by restaurant_name
order by count(*) desc
limit 10;
```

Fig: SQL Query for Jaipur City top restaurant Analysis

```
with join_cte1 as
(select price, rating, rating_count,full_date,year,month,month_name,quarter, day,week,state,city, location, restaurant_name,
category,dish_name
from fact_swiggy_orders f
join dim_date d on f.date_id=d.date_id
join dim_location l on f.location_id=l.location_id
join dim_restaurant r on f.restaurant_id=r.restaurant_id
join dim_category c on f.category_id =c.category_id
join dim_dish dd on f.dish_id=dd.dish_id
order by order_id )
select restaurant_name, count(*) from join_cte1 where state = 'Gujarat' and rating =5
group by restaurant_name
order by count(*) desc
limit 10;
```

Fig: SQL Query for 5 s Rated Restaurants in Gujarat